DAVID ADEOGUN

(770) 882-6032 | adeoguntdavid@gmail.com | Portfolio

PROFESSIONAL SUMMARY

Machine Learning Engineer with hands-on experience in model development, training, debugging, optimization, and evaluation across applied ML research and production systems. Proven track record of designing and deploying end-to-end ML pipelines across computer vision, natural language processing, time-series analysis, and robotics perception — from data generation to model deployment and monitoring. Skilled in benchmarking model performance, reasoning about ML system design tradeoffs, and communicating complex technical findings with clarity. Experienced in building scalable ML infrastructure on AWS (SageMaker, ECS, S3) and Kubernetes, with a demonstrated ability to bridge the gap between cutting-edge research and production-grade engineering.

TECHNICAL SKILLS

ML & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, LSTMs, CNNs, Vision Transformers, AdaBoost, YOLO, XGBoost, LightGBM, Random Forests, Neural Networks, ONNX Runtime, TensorFlow Serving, Hugging Face Transformers, Stable Diffusion, Reinforcement Learning

Experimentation & Evaluation: MLflow, Weights & Biases, Experiment Tracking, Model Registry, Hyperparameter Tuning, Optuna, Ray Tune, A/B Evaluation, Benchmark Design, RMSE/Precision/Recall/F1 Metrics, Cross-Validation, Feature Importance Analysis, SHAP

ML Infrastructure & Ops: AWS (SageMaker, ECS, ECR, S3, Lambda), Kubernetes, Docker, Apache Airflow, Kubeflow, CI/CD for ML Pipelines, GitHub Actions, Automated Rollbacks, Model Versioning, DVC, Feature Stores, Production Monitoring, Prometheus, Grafana

Programming & Data: Python, C++, SQL, MATLAB, NumPy, Pandas, Apache Kafka, MQTT, Spark, dbt, PostgreSQL, Redis, REST APIs, FastAPI

Perception & Robotics: OpenCV, PCL, ROS/ROS2, LiDAR Processing, 3D Point Cloud Segmentation, Sensor Fusion, SLAM, Computer Vision, Depth Estimation, Object Detection

INDUSTRY EXPERIENCE

Machine Learning Engineer | Mode Robots — Oct 2023 – Present

- Designed and trained temporal ML models including LSTMs and AdaBoost classifiers for predictive maintenance, achieving earlier fault detection with a 5,000+ cycle lead time, less than 5% false alarm rate, and RUL prediction RMSE under 2,000 cycles across time-series datasets with 150,000+ labeled cycles.
- Engineered feature extraction and selection pipelines across structured time-series and sensor data, applying dimensionality reduction techniques to improve model accuracy and reduce training time across high-dimensional IIoT telemetry datasets.
- Developed synthetic data generation pipelines with realistic noise injection and multiple degradation patterns — linear, exponential, and logarithmic — using NumPy and Pandas, creating controlled training and benchmarking environments for model robustness evaluation.
- Designed and evaluated CNN and Vision Transformer architectures for image-based perception tasks, benchmarking classification accuracy, inference speed, and model generalization across remote sensing and satellite imagery datasets.
- Fine-tuned and evaluated large language models for domain-specific reasoning tasks, designing schema-driven evaluation pipelines using AWS Bedrock to benchmark groundedness, decision-logic, and response quality across frontier models.
- Assessed and debugged AI-generated solutions across the full ML lifecycle, identifying failure modes in model training and optimization pipelines using PyTorch, scikit-learn, and ONNX Runtime to improve convergence and generalization.
- Benchmarked temporal vs. non-temporal architectures, conducting detailed error analysis on sequential data processing capabilities to inform production model selection and ML system design tradeoffs.
- Applied hyperparameter optimization using Optuna and Ray Tune across model training runs, improving model performance metrics including RMSE, precision, recall, and F1 scores across multiple production ML systems.
- Implemented model versioning, experiment tracking, and model registry management using MLflow and Weights & Biases, ensuring full reproducibility across training runs and deployment stages.
- Containerized ML models using Docker and deployed scalable inference endpoints on AWS ECS and Kubernetes, optimizing TensorFlow Serving and ONNX Runtime for production latency and throughput.

Cloud Platform Engineer | Natterbase — Feb 2022 – Sep 2023

- Designed and built secure, scalable AWS foundations including VPC architecture, IAM policies, and EC2 infrastructure, automating deployments using modern DevOps practices and managing EKS-based containerized workloads.

- Developed and maintained infrastructure-as-code using Terraform and AWS CloudFormation, streamlining cloud resource provisioning, reducing deployment time, and ensuring consistency across development and production environments.

Software Engineer | Williams Sonoma Inc Jun 2020 – Jan 2022

- Engineered and scaled backend services for catalog, cart, and checkout workflows, supporting high-traffic production systems and ensuring system reliability during peak loads

MACHINE LEARNING RESEARCH EXPERIENCE

Kennesaw State University – *Digital Automation Research Lab*

Advisor: Professor David Guerra-Zubiaga

- Trained and compared LSTM time-series model against AdaBoost baseline classifier for predictive maintenance, achieving earlier fault detection (5,000+ cycle lead time), <5% false alarm rate, and RUL prediction RMSE <2,000 cycles.
- Designed digital twin of ABB IRB 1200 robotic cell in Siemens Tecnomatix Process Simulate, validating IIoT-to-ML data pipeline architecture for Industry 4.0 predictive maintenance evaluation.
- Engineered synthetic IIoT telemetry generation pipeline producing 150,000+ labeled cycles with realistic noise injection and three degradation patterns (linear, exponential, logarithmic) for model training and benchmarking.
- Demonstrated that temporal modeling (LSTM on 500-cycle sequential windows) captures gradual degradation trends missed by snapshot-based approaches, informing ML system design tradeoffs for time-series tasks.

Georgia Southwestern State University – *Intelligent Systems Research*

Advisor: Professor Alexander Yemelyanov

- Architected AI-driven decision-support framework combining LLM reasoning, structured decision logic, and cognitive-behavioral problem-solving to produce explainable recommendations for risk-sensitive scenarios.
- Embedded Azure OpenAI (GPT-4o) as the core personalization engine, designing schema-driven user-profile pipelines to generate grounded, user-specific trade-offs and alternative rankings.
- Productionized and evaluated AI systems on Google Cloud with guardrails, telemetry-based quality monitoring, and cross-platform QA, delivering end-to-end architecture documentation.

PUBLICATIONS

[1] Yemelyanov, A., Adeogun, D. "AI-Powered Cognitive-Behavioral Decision Support for Risk Reduction in Insurance Coverage Selection." AHFE 2026. (Submitted)

PREPRINTS

[1] Adeogun, D., Vu, K., Alam, W., Quansah, E. "Digital Twin–Driven Predictive Maintenance for Pick-and-Place Robotic Cells Using IIoT Telemetry and Machine Learning." ASME IMECE 2026. (To Be Submitted) https://doi.org/10.13140/RG.2.2.15447.53926

[2] Adeogun, D., Sandoval, S., Wulfekuhle, V., Vu, K., Kouwenhoven, F. "Improving Efficiency of Deep-Learning based Vision Transformer for Remote Sensing for Satellite Images." ResearchGate, January 2026. https://doi.org/10.13140/RG.2.2.13764.49282

EDUCATION

M.S. Computer Science | Georgia Southwestern State University

- *Thesis: Design and Implementation of a User-Centered AI-Assisted Decision-Support Framework Under Risk and Uncertainty*

M.S. Intelligent Robotic Systems | Kennesaw State University

- *Relevant Coursework: Machine Learning, Machine Vision, Perception/Navigation/Path Planning, Robot Simulation, 3D Point Cloud Manipulation*

B.S. Software Development | Brigham Young University-Idaho